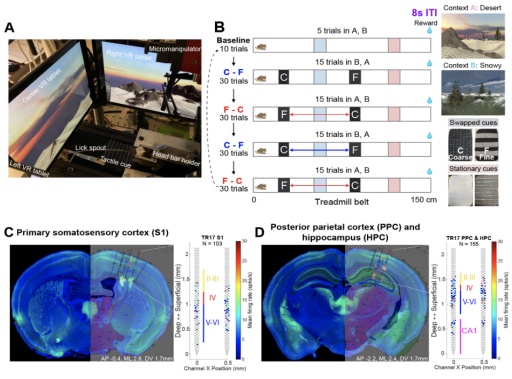
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Mice and Memory: An Analysis of Neural Signals and Hippocampal Activity during awake rest

I've always had a fundamental passion for neuroscience. I've long been told I have the mind of a researcher, having a deep love for the details, always asking more questions than anyone has the answers to, especially when it comes to the mind. Last summer I was finally given the chance to pursue some of those questions.

I had the opportunity to research the electrical signal occurring in the brain during awake rest in mice at the University of California Irvine's summer Institute for Neuroscience program. I was able to do this using electrophysiology techniques with the McNaughton lab. My work contributes to the project which used a novel VR treadmill set up to examine cortical dynamics in PPC (Posterior Parietal Cortex) and in S1 (Primary Somatosensory Cortex) during a visual and tactile task. Specifically, I was investigating if there are regional and temporal differences between context (ex. VR Biome) vs. content (ex. tactile cue) reactivation during awake rest.



A) Treadmill with VR setup for head-fixed mice. B) Behavioral task design. At the end of each trial, the mouse receives a reward followed by an 8 sec inter-trial interval (ITI) when the brake is on and VR tablets are black. C) Histology showing the 256F silicon probe recording locations (DiD dye) in S1 and in D) PPC and dorsal CA1, stained with DAPI, WFS1 (L2), and NECAB1 (L4) antibodies. Probe map shows the single unit yield across depth in each region, color coded by the mean firing rate.

Even though the research team collected an extensive amount of data, nobody has previously analyzed the spiking between laps. To address this gap in understanding the data, the treadmill was set to stop for 8 seconds after each lap, during which the VR went dark and the mouse was not standing on any tactile cue. I was able to participate in three successful recordings which followed three successful craniotomies, as well as some of the immunohistochemistry and staining process of processing the brains for data.

I learned to perform and was able to assist in every step of the experiment's pipeline. This included of training the mice, performing the craniotomies, prepping, painting, and inserting the probe, recording up to 10hrs of data, perfusing the mouse, extracting the brain, slicing it using a cryostat, selecting brain slices for processing, Immunohistochemistry and imaging.

However, I feel my largest contribution was the ability to analyze that un-analyzed data using Matlab. I started with a simple script that would create a raster plot, each spike would be represented as a dash on a 8 second time line, representing the 8 seconds of awake rest. This would then be repeated for each trial creating an image made up of hundreds of rows of dashes. This type of raster plot visualizes the spiking rate and patterns of each cell. In order to get a more complete picture, I eventually changed the time line to cover the two seconds before and after

rest, creating a figure spanning from -2 to 10 seconds. Once this script was complete, it had to be thoroughly debugged before I could make it loop through each cell, generating hundreds of figures for each animal.

I was able to see many different types of firing patterns and was able to identify some cells that had interesting ramping and decaying behaviors, reward signaling responses, and also delayed signaling that indicates it was not for the reward given at the beginning of those eight seconds, but rather something else. I identified a stripe-like pattern in some of the cells which indicated to me that there may be something lap or trial specific.
After observing this stripe-like pattern in multiple cells, I created a script that would display the laps in trial type groups rather than chronologically. The type trial that occurred right before the lap indicated the trial type by which the data was sorted.

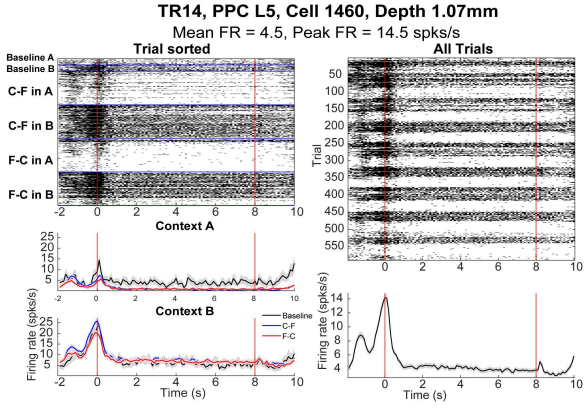


Fig 2: Raster plots and corresponding line graphs for a Cell found in Layer 5 of PPC. The right side graph shows the laps in chronological order, along with the average firing rate over time. A consistent strip pattern can be seen in the spiking. On the Left the trials have been sorted by trial type, which is defined by both tactile cue position and visual context, for a total of 6 trial types. It can be seen that the trials in context B, with C-F and F-C cue configuration, have a sustained higher firing rate than those in context A during rest. The reward signal also appears to be higher in context B trials. (TR14 = Treadmill mouse #14, PPC= Posterior Parietal Cortex, L5 = Layer 5, Cell 1460 = Cell ID number, Depth 1.07 = depth of the silicone probe measured from brain surface),(C= Coarse tactile cue, F= fine tactile cue, A = Desert visual context, B = Snowy visual context)

When sorted by trial type, these stripe patterns are revealed to correspond with specific trial types (fig.2). Specifically fig 2 shows sustained selective firing for visual context B with tactile cues during there 8 second of awake rest following the lap, during which the visual and tactile stimuli has been removed. The Baseline for context A also appears to have a higher firing rate, closer to that of the other trials in B, however due to the smaller number of baseline trials, I cannot be certain that it is not noise. Once the script used to make this figure was thoroughly debugged, I coded a loop that would generate and save sorted versions of each cell. I found that they were cells that were selectively firing for different tactile or visual stimuli, with some cells representing interactions between these two. I found the most context selective cells in PPC.

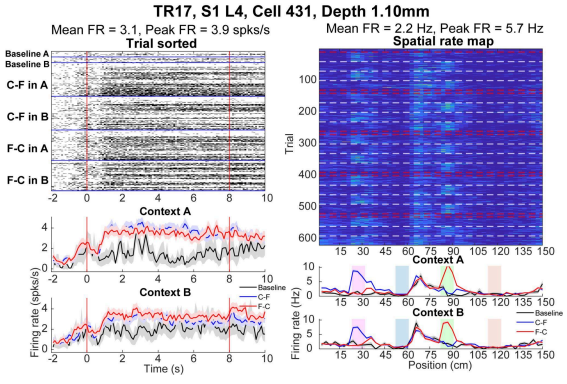


Fig 3. Raster plots and corresponding line graphs for a Cell found in Layer 4 of S1. The Left side graph shows the laps sorted by trial type, along with the average firing rate over time. The right shows the corresponding spatial rate map, with the trials in chronological order, showing firing relative to position on the 150cm treadmill. Vertical striping patterns can be seen

near 30cm and 90cm, aligning with the novel cue positions.
(TR17 = Treadmill mouse #17, S1= Primary Somatosensory Cortex, L4 = Layer 4, Cell 431 = Cell ID number, Depth 1.10mm = depth of the silicone probe measured from brain surface),(C= Coarse tactile cue, F= fine tactile cue, A = Desert visual context, B = Snowy visual context)

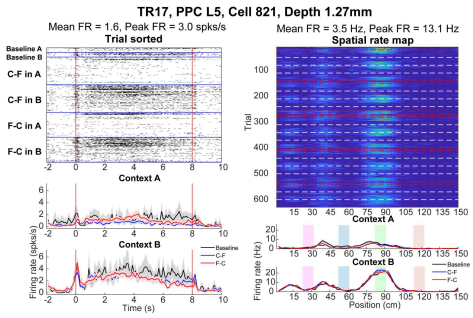


Fig 4. Raster plots and corresponding line graphs for a cell found in Layer 5 of PPC. The left side graph shows the laps sorted by trial type, along with the average firing rate over time. The right shows the corresponding spatial rate map, with the trials in chronological order, showing firing relative to position on the 150cm treadmill. A vertical striping pattern can be seen from approximately 75 cm to 90cm, aligning with a firing rate spike seen in context B.
(TR17 = Treadmill mouse #17, PPC= Primary Somatosensory Cortex, L5 = Layer 5, Cell 821 = Cell ID number, Depth 1.27mm = depth of the silicone probe measured from brain surface),(C= Coarse tactile cue, F= fine tactile cue, A = Desert visual context, B = Snowy visual context)

S1 had more cue selective cells. One stood out in particular for having higher firing rates for the trials with novel cues present (C-F or F-C) compared to the baseline trials. The trial sorted graph had some noise making this a bit unclear, so I decided to find the spatial rate map for the same cell. When put side by side I was able to see this cell was also distinctly cue selective in the spatial rate map, making it cue selective during both active stimuli and during rest. I also found certain context selective cells in PPC that were similarly selective on spatial rate maps (fig 3). Similarly, one cell fired selectively for context B over context A during the rest period, and the corresponding special rate map shows the same context selective behaviors, thus displaying reactivation (fig 4).

While most of my analysis was focused on S1 and PPC, I did also begin to investigate hippocampus ripples and their alignment to other firing behaviors. Preliminary results using that data of 3 animals indicated S1 narrow interneurons show peak firing activity ~0.35 sec after HPC ripple peak times. I intend to further research this area, adding more animals to the data pool when spike sorting and data processing is complete, and refining both the script and parameters based on existing literature.

Overall, what this data is able to do is confirm that memory is happening. This allows us to begin understanding the mechanism of memory reactivation in a healthy brain. It is infinitely more difficult to fix a system when we don't know how it's supposed to work in the first place. While basic science is often not as glamorous as disease models, it is my hope that through the fundamental understanding of memory that this research is building, we will be able to better understand when and how it breaks, and have baseline functioning that we can aim to restore.

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